M08 # Custom Agents

Methodology Intermediate

Create specialized sub-agents to delegate tasks

Agent structure

An agent is a Markdown file placed in `.claude/agents/` with a YAML frontmatter. Claude Code discovers it automatically at session startup.

```
---
name: code-reviewer
description: Use to review code before any commit
model: sonnet
tools: Read, Grep, Glob
---
```

The file body contains instructions in Markdown: role, methodology, examples, constraints.

Frontmatter fields

Field	Required	Role
`name`	Yes	kebab-case identifier
`description`	Yes	When Claude should delegate to it
`model`	No	Aliases such as `sonnet`, `inherit`, or a supported model ID
`tools`	No	Whitelist of accessible tools
`memory`	No	Persistent memory scope
`maxTurns`	No	Agentic turn limit
`permissionMode`	No	Permission control
`isolation`	No	`worktree` for git isolation
`color`	No	CLI color for visual distinction

Without `model`, a subagent inherits the parent model. `inherit` selects that behavior explicitly. Environment and organization overrides can still apply.

The tools field: isolation by default

Limiting `tools` to what the agent actually needs reduces the exposure surface and accidental errors. A code review agent only needs `Read, Grep, Glob`. No reason to give it `Bash` or `Write`.

Persistent memory (v2.1.33+)

```
memory: project   # .claude/agent-memory/<name>/
memory: user      # ~/.claude/agent-memory/<name>/
memory: local     # .claude/agent-memory-local/<name>/
```

`project` memory can be committed for team use; review it first. `local` memory stays private to the machine. Enabling agent memory automatically adds Read, Write and Edit, so omit `memory` for a strictly read-only agent.

Agent vs slash command

Criterion	Agent	Slash Command
Specialty	Yes, specific domain	No, generic workflow
Own memory	Yes (v2.1.33+)	No
Isolated tools	Yes, whitelist	No
Invocation	Automatic or manual	Manual only
Format	Markdown + frontmatter	Markdown template

Create an agent when the task recurs regularly with stable context and tools. Create a command when it is a one-off workflow or a procedure to follow.

Concrete example: code review agent

```
---
name: code-reviewer
description: Use PROACTIVELY after every code modification
  to verify quality, security, and conventions
model: sonnet
tools: Read, Grep, Glob
---

# Code Reviewer

Systematically verify: OWASP security, test coverage,
project convention compliance, and introduced technical debt.
```

The keyword `PROACTIVELY` in the description encourages Claude to invoke the agent without waiting for an explicit request.